함수의 극한

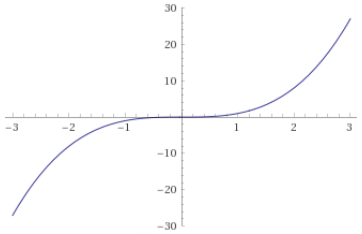
연습문제 풀이

02

연습문제 2.2

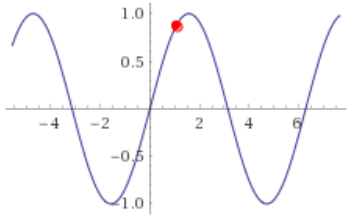
1. 다음 함수의 극한을 수치적인 방법과 그래프를 통한 방법으로 구하여라.

(1) $\lim_{x\to 2} x^3$



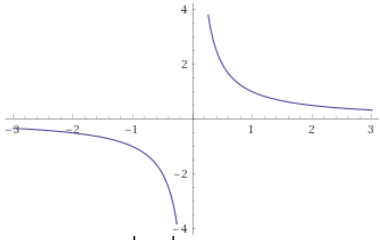
답: $\lim_{x\to 2} x^3 = 8$

(2) $\lim_{x\to \frac{\pi}{3}} (\sin x)$



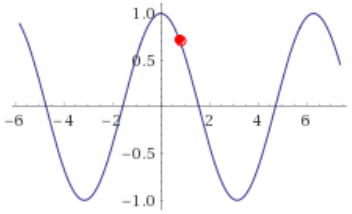
답: $\lim_{x\to \frac{\pi}{3}} (\sin x) = \frac{\sqrt{3}}{2}$

(3) $\lim_{x\to 2} \frac{1}{x}$



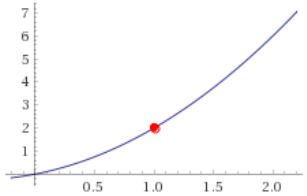
답: $\lim_{x\to 2} \frac{1}{x} = \frac{1}{2}$

(4) $\lim_{x\to \frac{\pi}{4}} \cos x$



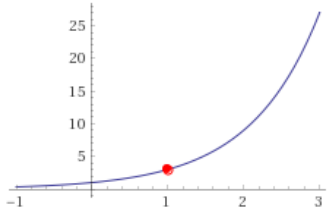
답: $\lim_{x\to \frac{\pi}{4}} \cos x = \frac{\sqrt{2}}{2}$

(5) $\lim_{x\to 1} (x^2+x)$



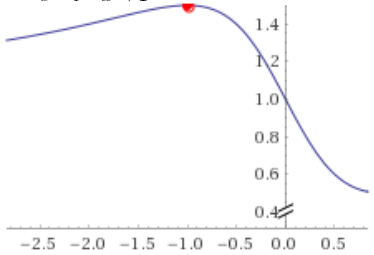
답: $\lim_{x\to 1} (x^2+x) = 2$

(6) $\lim_{x\to 1} 3^x$



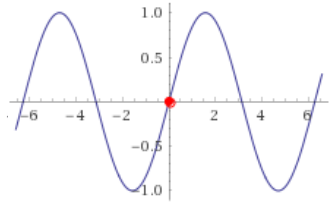
답: $\lim_{x\to 1} 3^x = 3$

(7) $\lim_{x\to -1} \frac{x^2-x+1}{x^2+1}$



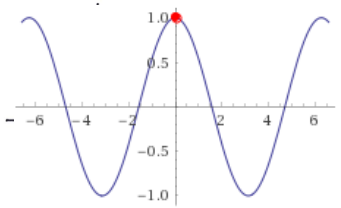
답: $\lim_{x\to -1} \frac{x^2-x+1}{x^2+1} = \frac{3}{2}$

(8) $\lim_{x\to 0} \sin x$



답: $\lim_{x\to 0} \sin x = 0$

(9) $\lim_{x\to 0} \cos x$



답: $\lim_{x\to 0} \cos x = 1$

연습문제 2.3

1. 다음 극한값을 구하여라.

(2) $\lim_{x\to 3}\sqrt{3x-5}$

$\lim_{x\to 3}\sqrt{3x-5}=\sqrt{9-5}=2$

답: 2

(3) $\lim_{x\to 2}\left(\frac{4x^3+8x}{x+4}\right)^{\frac{1}{3}}$

$\lim_{x\to 2}\left(\frac{4x^3+8x}{x+4}\right)^{\frac{1}{3}}=\left(\frac{32+16}{2+4}\right)^{\frac{1}{3}}=(8)^{\frac{1}{3}}=2$

답: 2

(4) $\lim_{x\to 4}\frac{x^2+2x-24}{x-4}$

$\lim_{x\to 4}\frac{(x+6)(x-4)}{x-4}=\lim_{x\to 4}x+6=10$

답: 10

2. $\lim_{x\to a}f(x)=3$, $\lim_{x\to a}g(x)=-1$ 일 때 다음을 구하여라.

(1) $\lim_{x\to a}\sqrt{f^2(x)+g^2(x)}$

$\lim_{x\to a}\sqrt{f^2(x)+g^2(x)}=\sqrt{3^2+(-1)^2}=\sqrt{10}$

답: $\sqrt{10}$

(2) $\lim_{x\to a}\sqrt[3]{g(x)}\,[f(x)+3]$

$\lim_{x\to a}\sqrt[3]{g(x)}\,[f(x)+3]=\sqrt[3]{-1}\cdot(3+3)=-6$

답: -6

(3) $\lim_{x\to a}[f(x)-(x-a)g(x)]$

$\lim_{x\to a}[f(x)-(x-a)g(x)]=3-\{(0)(-1)\}=3$

답: 3

3. $f(x)=\frac{1}{x}$ 일 때 $\lim_{x\to 2}\frac{f(x)-f(2)}{x-2}$ 를 구하여라.

$$\lim_{x\to 2}\frac{\frac{1}{x}-\frac{1}{2}}{x-2}=\lim_{x\to 2}\frac{\frac{-(x-2)}{2x}}{x-2}=\lim_{x\to 2}\frac{-(x-2)}{2x(x-2)}=\lim_{x\to 2}-\frac{1}{2x}=-\frac{1}{4}$$

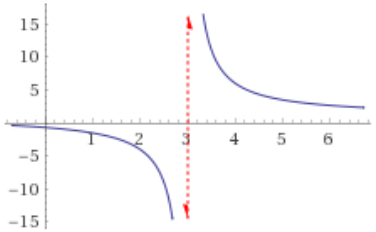
답: $-\frac{1}{4}$

연습문제 2.4

2. 다음 한쪽 극한값을 구하여라.

(1) $\lim_{x\to 3^+}\frac{x+2}{x-3}$

$\lim_{x\to 3^+}\frac{x+2}{x-3}=\lim_{x\to 3+}\frac{x-3+5}{x-3}=\lim_{x\to 3+}\left(1+\frac{5}{x-3}\right)=\infty$

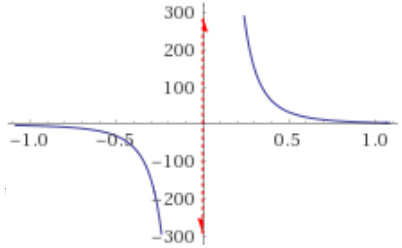


답: ∞

4. 다음 극한을 구하여라.

(1) $\lim_{x\to 0^-}\frac{4}{x^3}$

$\lim_{x\to 0^-}\frac{4}{x^3}=-\infty$



(4) $\lim_{x\to 0}\frac{x}{|x|}$

$i)\lim_{x\to 0+}\frac{x}{|x|}=\lim_{x\to 0+}\frac{x}{x}=1$

$ii)\lim_{x\to 0-}\frac{x}{|x|}=\lim_{x\to 0-}\frac{x}{-x}=-1$

답: $i)\neq ii)$이므로 존재하지 않는다.

(7) $\lim_{t\to 0}\frac{1-\cos t}{t^2}$

$\lim_{t\to 0}\frac{1-\cos t}{t^2}=\lim_{t\to 0}\left(\frac{1-\cos t}{t^2}\right)\left(\frac{1+\cos t}{1+\cos t}\right)=\lim_{t\to 0}\frac{\sin^2 t}{t^2(1+\cos t)}$

$(\because \sin^2 t+\cos^2 t=1)$

$=\lim_{t\to 0}\left(\frac{\sin t}{t}\right)^2\left(\frac{1}{1+\cos t}\right)=1\cdot\frac{1}{2}=\frac{1}{2}$

$(\because \lim_{t\to 0}\frac{\sin t}{t}=1,\ \cos 0=1)$

답: $\frac{1}{2}$

(8) $\lim_{x\to 0}\frac{\sqrt{1+x}-\sqrt{1-x}}{x}$

$\lim_{x\to 0}\frac{\sqrt{1+x}-\sqrt{1-x}}{x}=\lim_{x\to 0}\left(\frac{\sqrt{1+x}-\sqrt{1-x}}{x}\right)\left(\frac{\sqrt{1+x}+\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}}\right)$

$=\lim_{x\to 0}\frac{2x}{x(\sqrt{1+x}+\sqrt{1-x})}=\lim_{x\to 0}\frac{2}{\sqrt{1+x}+\sqrt{1-x}}=1$

답: 1

연습문제 2.5

1. 함수의 연속성을 조사하여라. 불연속이면 그 이유를 밝혀라.

(4) $g(t)=\dfrac{t^3-8}{t-2}$

$g(t)=\dfrac{t^3-8}{t-2}$는 $t=2$에서 불연속이다.

답: $t=2$에서 함수값 $g(2)$가 존재하지 않는다.

2. 다음 함수는 어떤 점에서 정의되지 않는다. 그 점에서 연속이 되게 하려면 어떻게 정의하여야 하는가?

(1) $f(x)=\dfrac{x^2-9}{x-3}$

$f(x)=\dfrac{x^2-9}{x-3}$는 $x=3$에서 정의되지 않는다

연속이 되게 하려면 $f(3)=\lim\limits_{x\to 3}f(x)=\lim\limits_{x\to 3}\dfrac{(x-3)(x+3)}{(x-3)}=6$

으로 정의하면 된다.

답 : 1) $x=3$에서 정의되지 않음

2) 연속이 되게 하려면 ? $\lim\limits_{x\to 3}f(x)=6$

3. 중간값 정리를 사용하여 $x^5+3x-3=0$의 근이 0과 1사이에 있음을 증명하여라.

$g(x)=x-f(x)$라 하면 $g(x)$는 $[0,1]$에서 연속이다.
또한 $g(0)=0-f(0)=-f(0)\le 0$이고 $g(1)=1-f(1)\ge 0$이므로($\because 0\le f(x)\le 1$)
중간값 정리(사이값 정리)에 의해서 $g(c)=0$을 만족하는 c가 0과1 사이에
적어도 하나 존재한다.
즉 $c-f(c)=0 \Rightarrow f(c)=c$